Does starting age predict who stays? Competition debut and long-term participation in youth track and field

Atle G. Guttormsen[1,*]
ORCID: 0000-0003-0188-8462

[1]School of Economics and Business, Norwegian University of Life Sciences (NMBU), Ås, Norway

*Corresponding author:
Atle G. Guttormsen
School of Economics and Business
Norwegian University of Life Sciences (NMBU)
P.O. Box 5003, 1432 Ås, Norway
E-mail: atle.guttormsen@nmbu.no

Running head: Debut age and retention in youth athletics

Word count: ∼3 000 (main text, excluding abstract, references, tables, and figure legends)

Abstract

Whether starting competitive sport early harms or protects long-term participation is contested: the early-specialisation literature cautions against early entry, while the largest prospective dropout study found that dropouts had started *later*. No population study has tested the question in an individual sport. Using the Norwegian Athletics Federation's complete competition registry, which is electronically complete from 2011, we followed every athlete born 2001–2005 whose first registered competition occurred at ages 10–16 (n = 17 496; 50 % female) to the end of 2026; 546 (3.1 %) were active as seniors (age $\geq$20). Senior conversion was U-shaped in debut age (non-linearity LR test $p < 0.001$): relative to debut at age 10 (3.8 %), adjusted odds were lowest for debuts at 11–14 (OR 0.53–0.75) and highest for debut at 16 (6.6 %; OR 1.73, 95 % CI 1.15–2.59), with a correspondingly null linear trend (OR 0.99 per year, 95 % CI 0.94–1.04). Measured from their own debut, however, later starters left much faster: five years after debut, 17 % of those who started at 10 were still competing against 3–7 % of those who started at 12–16, and survival curves with delayed entry converged by about age 16. Among athletes competing in the age-17 season, earlier debut appeared advantageous (OR 0.85 per year of later debut, 95 % CI 0.80–0.90), but the association disappeared once current competition volume was controlled (OR 0.99, 95 % CI 0.92–1.05). Neither the claim that early competitive entry undermines retention nor the claim that late entry does was supported. Starting age matters far less than current engagement — and every entry age produced senior athletes.

Keywords: youth sport; dropout; starting age; competition entry age; athlete retention; registry study; track and field

1 Introduction

When should children enter competitive sport? The question sits at the centre of the early-specialisation debate, and the two main positions make opposite predictions about participation. One line of argument, prominent in consensus statements, cautions that early competitive engagement risks burnout and withdrawal [1]; on this view, later entry should protect retention. The opposite prediction comes from the retention literature itself: in the largest prospective dropout study to date, among 1 762 Dutch football players, those who dropped out had *started organised football later* than those who stayed, and the authors recommended recruiting from early childhood [2] — though its exposure, joining organised football, is not the same construct as the registered-competition debut studied here, a gap we keep explicit throughout. Direct tests are scarce. Markers of early specialisation failed to predict actual dropout in the one study that measured both [3], systematic reviews of youth sport dropout identify motivational and social factors rather than starting age [4, 5], and the retrospective expertise literature — which finds that eventual elites often entered their main sport later [6, 7] — samples on success and therefore cannot speak to population-level retention. No study has followed a complete national entry population of an individual sport and asked what age at entry implies for who is still competing a decade later.

The Norwegian athletics registry permits exactly that design. Every sanctioned result in Norway is registered, the registry is electronically complete from 2011, and competition begins at age 10. For athletes born in 2001 or later, the full competitive career is therefore observed from its first day, with no left truncation: age at first registered competition is measured exactly, for everyone who ever entered, not only for those who later succeeded. Three earlier studies of the same registry frame the present one. Relative-age selection was found to be fully established at the entry stage and stable thereafter, locating the system's selection pressure at the point of entry rather than in adolescent talent identification [8]. A 14-year cohort study found that competition volume at ages 13–14 predicted senior retention far better than performance did, and that exit is preceded by behavioural withdrawal [9]. A companion study of performance trajectories found that progression predicts retention beyond volume — but that, here too, the *recent* window dominates the earlier one [10]. Against that background, entry age is the natural

next exposure: it is the first *registered* behavioural fact the registry holds about an athlete — prior training and other sports are invisible to it.

We pre-specified three questions. **H1:** among all athletes whose first registered competition fell at ages 10–16, is debut age associated with being active as a senior? Read onto this entry-timing margin, the two positions above suggest opposite directions, so H1 is two-sided, a U-shape is plausible, the categorical contrast is the primary display, and a likelihood-ratio test compares the categorical and linear codings. **H2:** once started, do later debutants exit faster, as the football findings suggest [2]? **H3:** among athletes still active at age 17, does entry history add anything beyond current engagement — or does the registry repeat its now-familiar lesson that current behaviour dominates history [9, 10]? Interpretation rules were fixed in advance: the exposure is *competition debut*, not the start of sport participation or training, and selection into late debut (already-interested adolescents choosing athletics) is expected and addressed in the design and interpretation. The analysis plan was committed to version control before any exposure–outcome model was estimated; the study was not externally preregistered.

2 Materials and methods

2.1 Data source and design

The Norwegian Athletics Federation's results database registers every sanctioned competition result in Norway and Norwegian athletes' results abroad. We used a frozen extract of 29 October 2026 (results dated on or before that day). Only participation fields were extracted — athlete, date, meet, club, approval status; no performance data of any kind were used at any age, in keeping with the purpose of the Norwegian children's sport regulations. Age is calendar-year age (competition year minus birth year), the basis of Norwegian age classes. Reporting follows STROBE [11].

2.2 Sample

All athletes born 2001–2008 entered the extraction (n = 33 666, with 675 707 result rows; 1 526 non-approved rows and 116 rows below age 10 were excluded). The 28 339 athletes with an

approved result at age 10 or older constitute the debutant population; 865 without registered sex (predominantly brief, early participants) and 732 (2.6 %) whose debut fell at age 17 or later (analysed as a sensitivity category) were excluded from the primary sample. Of the remaining 26 742 debutants, the primary sample comprises the 17 496 born 2001–2005 with debut at ages 10–16 — cohorts whose follow-up extends past age 20 for everyone — of whom 50.1 % were female and 546 (3.1 %) met the senior-activity outcome; the other 9 246, born 2006–2008, contribute to descriptive analyses of entry patterns only. Because non-approved rows and sub-age-10 rows were excluded before debut assignment, a handful of athletes may carry a slightly later registered debut than their true first appearance; the numbers involved are negligible. The full sample flow is given in Supplementary Table S1.

2.3 Exposure, outcomes, and covariates

The exposure is *debut age*: calendar-year age at the athlete's first approved registered result (10–16), treated categorically (reference: debut at 10) and as a linear trend. The primary outcome, identical to the companion studies [9, 10], is *active senior*: at least two approved results in the same calendar year at age 20 or older. Secondary outcomes were at least one result at age $\geq$20, activity at age $\geq$17, and retention k years after debut (k = 1–5), defined as at least one approved result in the calendar year debut + k (point prevalence), computed with per-k denominators over athlete-years fully observable by 2025. Covariates were sex, birth cohort, region (modal competition district in the debut season; 18 districts plus unknown), baseline club (modal club in the debut season) with the logarithm of a club-size proxy (study athletes per club), birth quarter where an exact birth date existed, debut-season competition days, and a COVID-19 flag for debuts in 2020–2021. Covariate sets per hypothesis: the primary H1 models adjust for sex, cohort, region, and club size, with the COVID-19 flag and debut-season days added in robustness variants; the age-scale Cox uses sex, centred cohort, and club size; the landmark models use sex and cohort, with current competition days added in the second specification. Full coefficient tables are available in the supplementary material.

2.4 Statistical analysis

H1 was estimated by logistic regression in the primary sample, with debut age categorical and linear, adjusted for sex, cohort, region, and club size, with club-clustered standard errors as a sensitivity. H2 was addressed two ways, because they answer different forms of the question. First, descriptively on the *duration* clock: the proportion still competing k years after their own debut, by debut age — the comparison that aligns athletes at the same career stage. Second, on the *age* clock: a Cox model with age as the timescale and delayed entry at debut [12], athletes leaving the risk set at their last active age (+0.5) and censored if active in 2025 or 2026. An exit therefore requires at least two fully idle calendar years for athletes last active in 2024 or earlier, and returns after 2026 are unknowable; the 2026 season is observed to 29 October, which slightly shortens the senior window of the youngest cohort — the cohort covariate and the cohort-split robustness address this. The age-scale coefficient on debut age compares, at the same age, athletes at different career stages; it therefore blends duration and entry effects, and is reported with that caveat, with the duration-clock table as the primary evidence for H2. H3 was a landmark analysis [13] among athletes with at least one approved result *in the age-17 season itself* — not merely any result at age 17 or later, which would let future senior activity qualify athletes into the landmark and thereby condition on the outcome's own future. Senior activity was regressed on debut age with sex and cohort, with and without current engagement (competition days at ages 16–17, standardised); the landmark sits at the end of the age-17 season, all predictors are measured by then, and the outcome window begins at age 20. Sex-specific conversion and a sex $\times$ debut-age interaction were also examined. Pre-specified robustness analyses excluded COVID-19 debuts, varied the outcome, added debut-season volume, split the cohorts (2001–2003 vs. 2004–2005), included debut at 17+ as a category, and explored a debut age $\times$ birth-quarter interaction — the last on the selected subsample with exact birth dates, since coverage rises steeply with debut age (Table 1). Analyses used Python 3.13 (pandas, statsmodels, lifelines).

2.5 Ethics and transparency

The data are public competition results; no individual is identified; no performance data were used, and nothing below age 13 beyond participation itself was analysed. As for the companion registry studies, the project falls outside the scope of the Norwegian Health Research Act, and an assessment from Sikt was obtained. The analysis plan, hypotheses, estimands, and interpretation rules were committed to version control before estimation; code and aggregate outputs are available, and athlete-level files are regenerable from the registry.

3 Results

3.1 Who debuts when

Debut at 10, the first eligible age, was the single most common entry point and its share grew across cohorts (32.6 % of debutants born 2001; 43.9 % born 2008), but late entry remained substantial: roughly one debutant in six entered at 14 or later (Figure 1). Table 1 describes the three debut groups. Late debutants were more often male (female share 52.2 % for debut at 10–11 against 42.0 % for 14–16). The typical debut was brief in every group: the median debutant competed on a single day in the debut season, and median career length was two seasons for debut at 10–11 and one season otherwise. Entry intensity was thus flat across debut ages — debut age is not a proxy for dose at entry. Exact birth dates — needed for birth-quarter analyses — existed for 38.6 % of early, 63.8 % of mid, and 88.5 % of late debutants, the registry's known pattern of missingness concentrating among brief early participants [8]; among those with known dates, the familiar first-quarter overrepresentation was present at every debut age (Q1 26–29 % against Q4 20–22 %).

Table 1: Characteristics of the primary sample (born 2001–2005, debut at ages 10–16), by debut group.

Characteristic	Debut 10–11	Debut 12–13	Debut 14–16
n	10 242	4 220	3 034
Female, %	52.2	50.8	42.0
Exact birth date available, %	38.6	63.8	88.5
Born Q1, % of those with birth date	29.2	26.1	27.1
Born Q4, % of those with birth date	19.9	21.9	19.8
Competition days in debut season, median (IQR)	1 (1–3)	1 (1–3)	1 (1–2)
Career length, seasons, median (IQR)	2 (1–4)	1 (1–3)	1 (1–2)
Debut season 2020–2021, %	0.0	0.0	6.3

By construction, cohorts born 2001–2005 could only experience a 2020–2021 debut at ages 14 and above.

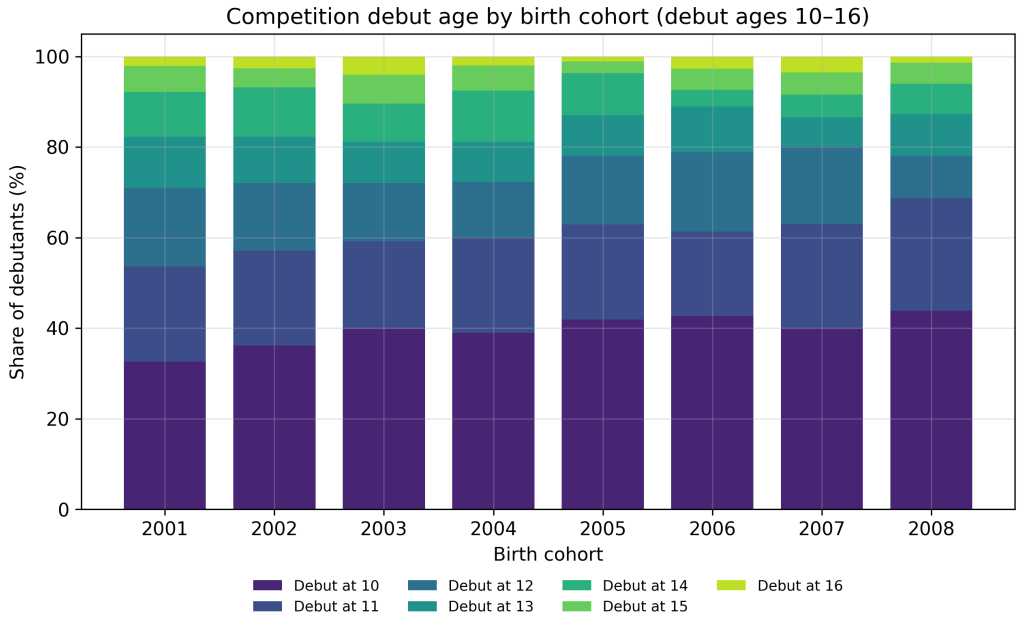


Figure 1: Competition debut age by birth cohort (all debutants born 2001–2008, debut ages 10–16).

3.2 Senior conversion is U-shaped in debut age (H1)

Of the 17 496 debutants, 546 (3.1 %) were active as seniors. Conversion was not monotone in debut age (Figure 2): it was 3.8 % for debut at 10 (253/6 626), fell to 2.4 % at 11, 2.1 % at 12,

2.9 % at 13 and 2.4 % at 14, and rose to 3.9 % at 15 and 6.6 % at 16 (28/424). A likelihood-ratio test rejected the linear coding in favour of the categorical one (LR = 43.2, df = 5, p = 3.3×10^{-8}): the U is formal, not visual. Adjusted for sex, cohort, region, and club size (Table 2), debuts at 11, 12 and 14 had significantly lower odds than debut at 10 (OR 0.53–0.63), debut at 13 directionally lower odds (OR 0.75, 95 % CI 0.55–1.02), debut at 15 the same odds (OR 1.00), and debut at 16 markedly higher odds (OR 1.73, 95 % CI 1.15–2.59, resting on 28 events). The linear trend across 10–16 was correspondingly null (OR 0.99 per year, 95 % CI 0.94–1.04) — a flat average concealing a U. Neither of the two simple expectations — early entry harms, or late entry harms — fits this shape. Conversion was higher for boys than girls at every debut age (4.3 vs. 3.4 % at debut 10; 7.1 vs. 5.7 % at debut 16; Supplementary Table S3), and the late-entry channel is male-tilted in composition (42.0 % female at debut 14–16); the sex $\times$ debut-age interaction was directionally negative but not significant (OR 0.92 per year, 95 % CI 0.82–1.02), so the U describes both sexes.

Table 2: Adjusted odds of senior activity (age $\geq$20) by debut age (n = 17 496; 546 events). Reference: debut at age 10. Adjusted for sex, birth cohort, region, and club size. A likelihood-ratio test favours the categorical over the linear coding (LR = 43.2, df = 5, $p < 0.001$); the debut-16 estimate rests on 28 events.

Debut age	OR (95 % CI)
10 (reference; raw conversion 3.8 %)	1.00
11	0.63 (0.49–0.81)
12	0.53 (0.39–0.71)
13	0.75 (0.55–1.02)
14	0.62 (0.44–0.86)
15	1.00 (0.70–1.45)
16	1.73 (1.15–2.59)
Linear trend, per +1 year of debut age	0.99 (0.94–1.04)

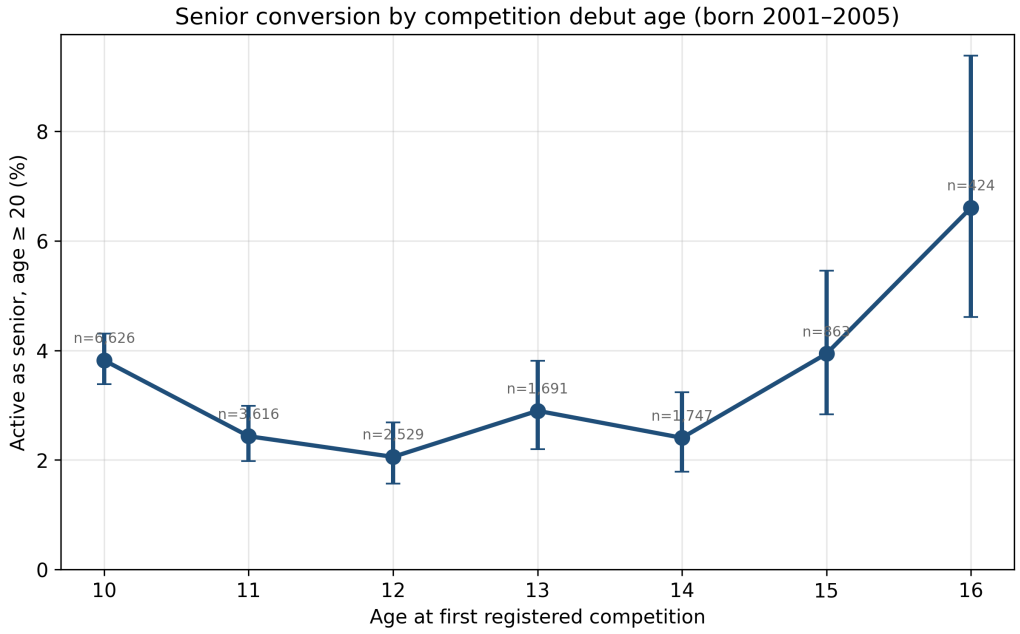


Figure 2: Senior conversion by competition debut age, with Wilson 95 % confidence intervals (born 2001–2005).

3.3 Later starters leave faster — on their own clock (H2)

Aligned on time since debut (Table 3; Figure 4), early entrants were far more durable. One year after the debut season, 51 % of those who debuted at 10 were still competing, against 34–40 % of those who debuted at 11–14, 26 % at 15, and 35 % at 16 — a reversal at the latest entry age that itself fits the self-selection account developed below. Five years after debut the gap had widened to 17 % (debut 10) and 10 % (debut 11) against 3–7 % (debuts 12–16). Denominators are fully observable athlete-years and shrink only for debut at 16 (424 at k = 1 to 389 at k = 5); Wilson intervals for every cell are given in Supplementary Table S2. On the age clock, Kaplan–Meier curves with delayed entry (Figure 3) show every debut group plunging through its own first seasons — and then converging: from about age 16 onward the three groups' survival ran essentially together through the mid-twenties. The age-scale Cox model summarises the same pattern (HR 1.32 per year of later debut, 95 % CI 1.31–1.34), but this coefficient compares athletes at the same age across different career stages and so blends entry effects with the universal early-career churn; the duration-aligned comparison above is the evidentially cleaner statement. Sex was not associated with exit hazard in this population-wide

model (HR 1.01, 95 % CI 0.98–1.04).

Table 3: Proportion still competing k years after the debut season (at least one approved result in the calendar year debut + k), by debut age (born 2001–2005; athlete-years fully observable by 2025). Denominators equal the stated n except for debut at 16, where they shrink from 424 ($k \leq 4$) to 389 (k = 5); Wilson 95 % intervals for every cell are given in Supplementary Table S2.

Debut age	n	k=1	k=2	k=3	k=4	k=5
10	6 626	0.51	0.38	0.29	0.22	0.17
11	3 616	0.40	0.27	0.19	0.13	0.10
12	2 529	0.37	0.23	0.14	0.09	0.07
13	1 691	0.38	0.23	0.15	0.10	0.07
14	1 747	0.34	0.13	0.08	0.06	0.03
15	863	0.26	0.13	0.08	0.05	0.04
16	424	0.35	0.21	0.11	0.07	0.06

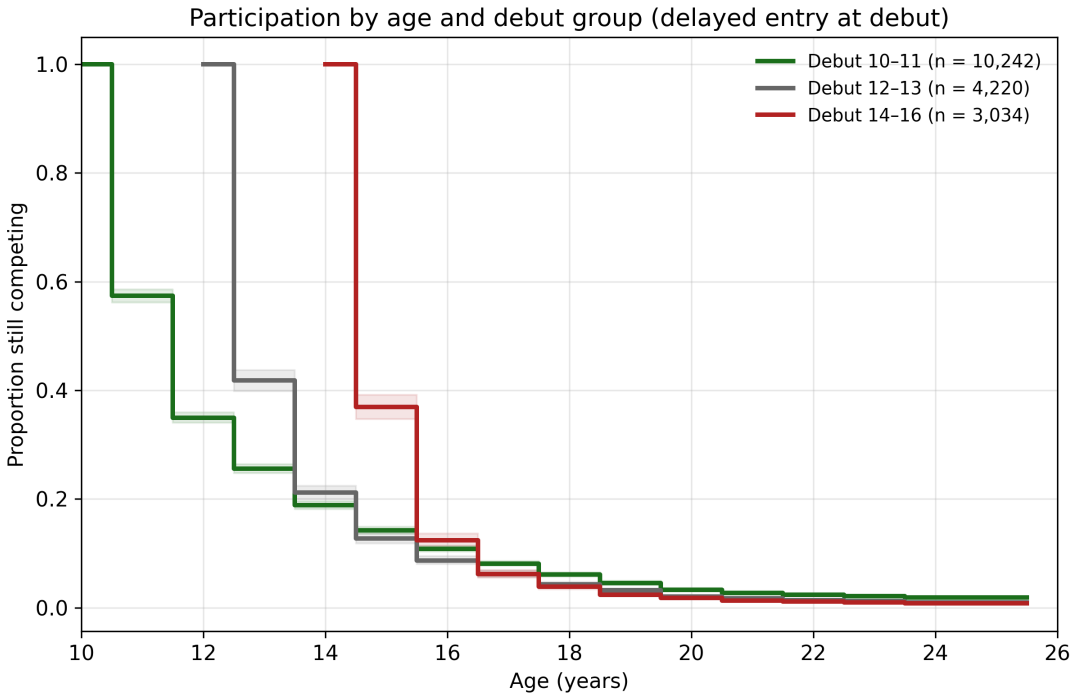


Figure 3: Participation by age and debut group, Kaplan–Meier with delayed entry at debut (born 2001–2005). Athletes active in 2025–2026 are censored. Curve segments before all groups have entered describe only the groups already at risk. The curves converge from about age 16.

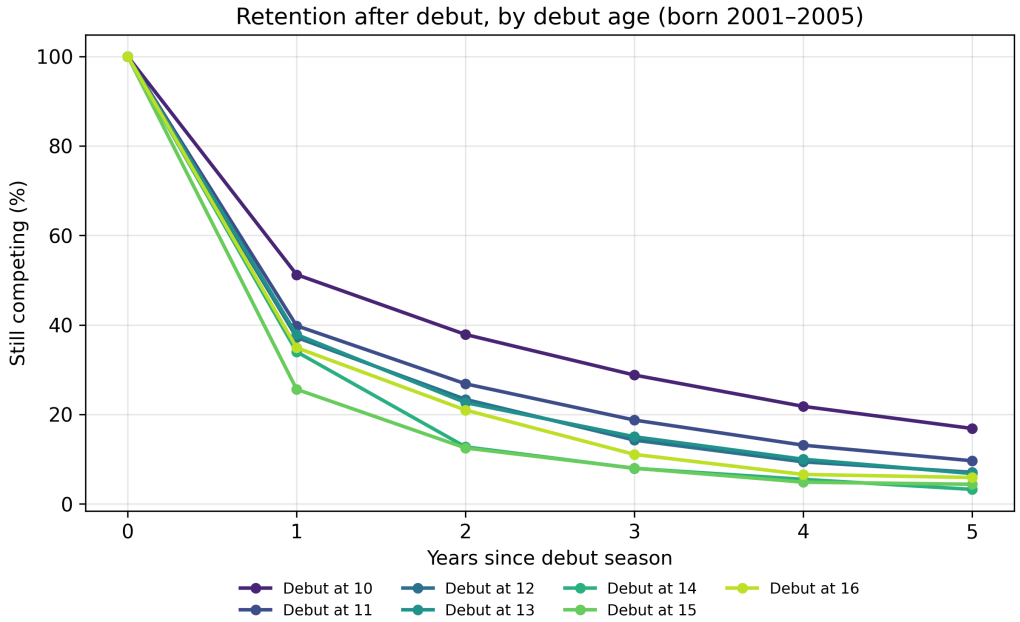


Figure 4: Retention k years after the debut season, by debut age (born 2001–2005).

3.4 Among survivors, history adds nothing once current engagement is known (H3)

Of the 17 496 debutants, 1 637 (9.4 %) competed in the age-17 season itself; 472 of these (28.8 %) became active seniors, while 74 seniors had a fully idle age-17 season before returning — which is exactly why the landmark must be defined on the season itself rather than on any activity at age 17 or later. In this landmark sample, earlier debut appeared advantageous: OR 0.85 per year of later debut (95 % CI 0.80–0.90). Adding current engagement — competition days at ages 16–17 — removed the association entirely (OR 0.99, 95 % CI 0.92–1.05). Conditional on competing at 17 and on how much the athlete competes now, when they started no longer carries predictive information. Two readings must nevertheless be kept apart: current volume is plausibly a *mediator* of entry timing, and conditioning on survival to the landmark invites index-event selection, so the result licenses no claim that entry history is causally inert — only that it adds nothing to prediction at the landmark. This replicates, for entry history, what the companion studies found for early competition volume [9] and early performance [10]: in this registry, recent behaviour dominates history.

3.5 Robustness

The pre-specified variations (Table 4) altered nothing material. Excluding COVID-era debuts left the trend null (0.95, 95 % CI 0.90–1.01); the one-result senior outcome gave 1.00; activity at 17 among debuts at 10–14 gave 0.97; conditioning on debut-season volume gave 0.94 (95 % CI 0.90–1.00); the cohort halves agreed (0.98 and 0.99); and athletes debuting at 17–19 — outside the primary range, and now excluding the 165 athletes whose first registered result came at age 20 or later and could therefore coincide with the outcome itself — converted at distinctly higher rates (OR 3.36 vs. debut at 10, 95 % CI 2.42–4.67; within the late group, 193 debuted at 17, 153 at 18 and 56 at 19), an extension of the U's right arm consistent with strong self-selection into very late entry. The exploratory debut age $\times$ birth-quarter interaction was compatible with no modification (OR 1.04 per quarter-year, 95 % CI 1.00–1.09), on the selected 53 % of the sample with exact birth dates.

Table 4: Robustness analyses. Estimates are for debut age (per +1 year, linear) unless stated; outcome is senior activity unless stated.

Analysis	n	Events	OR (95 % CI)
Primary linear trend	17 496	546	0.99 (0.94–1.04)
Excluding debuts in 2020–2021	17 305	530	0.95 (0.90–1.01)
Outcome: $\geq$1 result at age $\geq$20	17 496	694	1.00 (0.96–1.05)
Outcome: any result at age $\geq$17 (debuts 10–14)	16 209	1 621	0.97 (0.93–1.00)
Adding debut-season competition days	17 496	546	0.94 (0.90–1.00)
Cohorts born 2001–2003	10 667	349	0.98 (0.92–1.05)
Cohorts born 2004–2005	6 829	197	0.99 (0.91–1.08)
Debut at 17–19 vs. debut at 10 (categorical)	17 898	595	3.36 (2.42–4.67)
Adding the COVID-19 debut flag (linear trend)	17 496	546	0.95 (0.90–1.01)
Debut age $\times$ birth quarter (exploratory)	9 330	538	1.04 (1.00–1.09)
Club-clustered robust SE (trend)	17 496	546	0.99 (0.93–1.04)

4 Discussion

4.1 Principal findings

In the complete entry population of Norwegian youth track and field, age at first competition carried no simple message about who would still be competing as an adult. Senior conversion was U-shaped — highest for the earliest (10) and the latest (16, and 17+) entrants, lowest in between — with a null linear trend. Measured from their own start, later entrants churned much faster, half of the ten-year-old debutants surviving the first year against roughly a quarter to a third of those debuting at 14–16; yet survival curves converged by about age 16, and among athletes competing in the age-17 season, entry history carried no predictive information once current competition volume was known.

4.2 What the entry-timing margin can and cannot test

Neither of the positions that motivated the study was formulated about this exposure, and the revision of both must respect that. The specialisation cautions concern specialised, high-volume, year-round involvement [1] — not whether a ten-year-old appears once in a sanctioned meet (the median debutant competed on a single day); the football finding concerns joining organised sport [2], whereas a "late debutant" here may have trained for years before a first meet. What this study tests directly is *entry timing into registered competition* — an administrable, policy-relevant variable federations actually control records for — and it comments on the specialisation debate only indirectly. On that margin, the data support neither simple expectation. Early debutants did not convert worse — they converted better than the 11–14 entrants and were by far the most durable on the duration clock — and late entrants did not simply fail: those who entered at 15–16, and especially 17–19, converted at the highest rates of all.

The resolution we offer is selection, with different candidate mechanisms on the two arms. On the right arm, entry at 15 and later is plausibly a decision adolescents make for themselves, and those who make it are a small, motivated, male-tilted group whose careers, when they fail, fail fast. On the left arm, several non-exclusive mechanisms could make debut at 10 beat 11–14: families with the strongest athletics attachment enrol at the first eligible age; mid-childhood entrants may be samplers arriving from other activities with weaker athletics identity; and club pathways built around training groups formed at 10 may serve later joiners less well. We present the parental-decision reading of early entry as interpretation, not evidence. And the fullest honesty is this: the entire U may be compositional — debut age may carry no behavioural effect at all, only information about who enters when. The registry cannot see motivation, prior training, or participation in other sports, and competition debut is not the start of sport involvement.

4.3 The first season is the universal filter

The dominant feature of every debut group is the first-year cliff: the median debutant competed on one day, and between half and three quarters of each entry cohort never established themselves. This matches, at the behavioural level, the entry-stage account of the registry's

relative-age structure — selection into sustained participation happens at and immediately after entry, not in adolescent talent identification [8] — and it extends it beyond birth-date composition to participation as such. Whatever determines long-term engagement, most of its action is over within a season or two of arrival, at every entry age.

4.4 History versus current engagement, third instalment

The landmark result completes a pattern now replicated three times in this registry with three different exposures. Early competition volume predicts senior retention, but later volume dominates it [9]. Early performance progression predicts retention, but through the most recent window [10]. And entry age predicts nothing among 17-year-old survivors once current volume is in the model. For talent systems, the registry keeps returning one lesson: *as a predictor*, what an athlete is doing now outperforms any summary of how they got there. The claim is strictly predictive. Current engagement is plausibly a mediator of entry history, and the landmark conditions on survival to 17, so the attenuation of the debut coefficient is what a mediated exposure would show even if entry timing mattered causally. What follows for practice is correspondingly modest: at the landmark, entry-age heuristics add nothing to what current participation already says, in either direction.

4.5 Strengths and limitations

The strengths are completeness and design: a national registry with no left truncation for these cohorts, exact entry measurement for everyone who ever entered (not only later successes, the known weakness of retrospective expertise samples [6, 7]), explicit delayed-entry survival methods, a plan committed before estimation, and internal replication across cohort halves. The limitations are equally structural. Competition debut is not sport start: children may train for years before their first meet, and late debutants may arrive from other sports with long athletic histories, so the exposure is entry into *registered competition*. Exit from the registry is not exit from sport. Selection into debut age is unmeasured and surely strong; no causal reading is offered. Birth dates are missing for most brief early participants, so the birth-quarter analysis is exploratory and selected. Registered debut also reflects registration practice — which

meets submit results for the youngest classes, and how clubs handle them — so cohort trends in debut age (Figure 1) may be partly organisational rather than behavioural. Selection into debut age plausibly tracks geography, club culture, family resources, and socioeconomic access, none of which the registry measures, and the club-size proxy is defined within the study sample. Retention is not attainment: no performance data were used, so nothing here says where elite performers come from — the retrospective expertise literature cannot speak to retention, and this study cannot speak to attainment. The senior outcome is rare (3.1 %), so even twofold relative differences are small in absolute terms. Finally, the age-scale Cox estimate blends entry and career-stage effects, which is why the duration-aligned retention table carries the H2 conclusion.

5 Perspective

For federations and clubs, these population figures provide three descriptive planning parameters. First, there is no wrong age to start: every entry age from 10 to 19 produced senior athletes, and the latest entrants converted at the highest per-head rates. These are yields of *self-selected* entrants: adolescents recruited into entry who would not have come on their own cannot be assumed to convert at the observed 6.6 %, and the late channel as observed is male-tilted, which matters for what a recruitment drive would actually reach. Second, the decisive window is the first season or two after entry, whenever entry happens: between half and three quarters of every entry cohort resolves within two seasons, so retention effort belongs at the entry transition rather than in managing entry timing — a conclusion that does not depend on the selection caveat. Third, by late adolescence the entry groups are indistinguishable: survival curves converge from about age 16, and among those competing at 17, current engagement carries all the predictive information. All three statements concern participation in registered athletics, not sport participation or physical activity in general.

Declarations

Funding. The author received no funding for this work.

Competing interests. The author declares no competing interests. The author maintains an open, non-commercial statistics platform built on the federation's public results database.

Ethics. The study analyses public competition results and falls outside the scope of the Norwegian Health Research Act; an assessment from Sikt was obtained. No individual athlete is identified. No performance data were used, and no analysis below age 13 goes beyond participation itself, in keeping with the Norwegian children's sport regulations.

Data availability. The underlying data are the Norwegian Athletics Federation's public competition results. Aggregate outputs and all analysis code are available from the author; athlete-level files are regenerable from the registry with the deposited code.

Author contributions. AGG conceived the study, performed all analyses, and wrote the manuscript.

Declaration of AI use. The author used a large language model (Claude, Anthropic) for programming assistance and manuscript drafting support. The author reviewed, verified, and edited all content and takes full responsibility for the integrity of the work.

Acknowledgements. The author thanks the Norwegian Athletics Federation, whose systematic registration of competition results makes population research of this kind possible.

Supplementary material

Table 5: *

Supplementary Table S1. Sample flow.

Step	n
Athletes born 2001–2008 in the registry extract	33 666
Result rows extracted (2011 – 29 Oct 2026)	675 707
excluded: non-approved rows	1 526
excluded: rows below age 10	116
Debutants with an approved result at age $\geq$10	28 339
excluded: missing sex	865
set aside: debut at age $\geq$17 (sensitivity analyses)	732
Debutants at ages 10–16 with known sex	26 742
primary sample: born 2001–2005	17 496
descriptive only: born 2006–2008	9 246
Active seniors in the primary sample	546

Table 6: *

Supplementary Table S2. Retention k years after the debut season: proportion (Wilson 95 % CI). Denominators equal the debut-age n except debut 16 at k = 5 (n = 389).

Debut age	n	k=1	k=2	k=3	k=4	k=5
10	6 626	.512 (.500–.524)	.379 (.367–.390)	.288 (.277–.299)	.218 (.208–.228)	.169 (.160–.178)
11	3 616	.398 (.382–.414)	.269 (.254–.283)	.188 (.175–.201)	.131 (.121–.143)	.097 (.087–.107)
12	2 529	.372 (.354–.391)	.234 (.218–.251)	.143 (.130–.157)	.094 (.083–.106)	.071 (.061–.081)
13	1 691	.379 (.356–.402)	.227 (.207–.247)	.150 (.134–.168)	.100 (.087–.115)	.068 (.057–.081)
14	1 747	.340 (.318–.363)	.128 (.113–.144)	.080 (.068–.093)	.055 (.045–.067)	.033 (.025–.042)
15	863	.256 (.228–.286)	.125 (.105–.149)	.080 (.064–.100)	.049 (.036–.065)	.044 (.032–.060)
16	424	.349 (.305–.396)	.210 (.174–.251)	.111 (.084–.144)	.066 (.046–.094)	.059 (.040–.087)

Table 7: *

Supplementary Table S3. Senior conversion by debut age and sex (born 2001–2005), % (Wilson 95 % CI).

Debut age	Girls, n	Girls, % (CI)	Boys, n	Boys, % (CI)
10	3 485	3.4 (2.8–4.0)	3 141	4.3 (3.6–5.1)
11	1 860	2.2 (1.6–2.9)	1 756	2.7 (2.1–3.6)
12	1 263	1.8 (1.2–2.7)	1 266	2.3 (1.6–3.3)
13	879	2.0 (1.3–3.2)	812	3.8 (2.7–5.4)
14	789	2.0 (1.3–3.3)	958	2.7 (1.9–3.9)
15	326	2.5 (1.2–4.8)	537	4.8 (3.3–7.0)
16	158	5.7 (3.0–10.5)	266	7.1 (4.6–10.9)

Sex × debut-age interaction (linear): OR 0.92 per year for girls relative to boys, 95 % CI 0.82–1.02. Full coefficient tables for the primary models are available with the analysis code.

References

[1] John P. DiFiori, Holly J. Benjamin, Joel S. Brenner, Andrew Gregory, Neeru Jayanthi, Greg L. Landry, and Anthony Luke. Overuse injuries and burnout in youth sports: a position statement from the American Medical Society for Sports Medicine. *Clinical Journal of Sport Medicine*, 24(1):3–20, 2014.

[2] Nico W. Van Yperen, Laura Jonker, and Jan Verbeek. Predicting dropout from organized football: a prospective 4-year study among adolescent and young adult football players. *Frontiers in Sports and Active Living*, 3:752884, 2022. Published online 17 January 2022; the DOI string carries the 2021 volume year.

[3] Heather K. Larson, Bradley W. Young, Tara-Leigh F. McHugh, and Wendy M. Rodgers. Markers of early specialization and their relationships with burnout and dropout in swimming. *Journal of Sport and Exercise Psychology*, 41(1):46–54, 2019.

[4] Jeff Crane and Viviene Temple. A systematic review of dropout from organized sport among children and youth. *European Physical Education Review*, 21(1):114–131, 2015.

[5] Lars Erik Espedalen and Ørnulf Seippel. Dropout and social inequality: young people's reasons for leaving organized sports. *Annals of Leisure Research*, 27(2):197–214, 2022.

[6] Michael Barth, Arne Güllich, Brooke N. Macnamara, and David Z. Hambrick. Predictors of junior versus senior elite performance are opposite: a systematic review and meta-analysis of participation patterns. *Sports Medicine*, 52(6):1399–1416, 2022.

[7] Arne Güllich, Michael Barth, David Z. Hambrick, and Brooke N. Macnamara. Recent discoveries on the acquisition of the highest levels of human performance. *Science*, 390(6779):eadt7790, 2025.

[8] Atle G. Guttormsen. Dropped out before they ever competed: a population-wide registry analysis of relative age effects in Norwegian track and field from age 10 to 25. *Sports Medicine – Open*, 2026. In press.

[9] Atle G. Guttormsen. Pulling back before dropping out: declining competition participation precedes exit from Norwegian youth track and field — a 14-year register study. Under review, 2026.

[10] Atle G. Guttormsen. Stagnation before exit: performance progression and retention in youth track and field. Manuscript in preparation, 2026.

[11] Erik von Elm, Douglas G. Altman, Matthias Egger, Stuart J. Pocock, Peter C. Gøtzsche, and Jan P. Vandenbroucke. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *The Lancet*, 370(9596):1453–1457, 2007.

[12] Edward L. Korn, Barry I. Graubard, and Douglas Midthune. Time-to-event analysis of longitudinal follow-up of a survey: choice of the time-scale. *American Journal of Epidemiology*, 145(1):72–80, 1997.

[13] James R. Anderson, Kevin C. Cain, and Richard D. Gelber. Analysis of survival by tumor response. *Journal of Clinical Oncology*, 1(11):710–719, 1983.